Autoregressive Language Models as Bayesian Optimal Controllers over Hidden Markovian State Spaces

A Unified Theoretical Framework

A Theoretical Synthesis

April 2026

Abstract

We develop a unified mathematical framework for understanding large autoregressive transformer language models through the lens of three interconnected formalisms: *hidden Markov processes*, *stochastic optimal control*, and *Bayesian inference*. We formalise the claim that the internal hidden state of a transformer defines a high-dimensional, continuous-state hidden Markov process whose effective memory decays exponentially, yielding asymptotic memorylessness governed by the spectral properties of an induced transfer operator. We then show that autoregressive token generation can be cast as a stochastic optimal control problem in which the model's hidden state serves as the control state, token emissions are control actions, and the learned value function encodes a Bellman-optimal policy over sequence trajectories. Finally, we demonstrate that the pretrained weights define a prior measure over token sequences, reinforcement learning from human feedback (RLHF) reshapes this prior into a value-weighted posterior, and in-context conditioning performs amortised variational Bayesian updating within each conversation. The three perspectives are proven to be mutually consistent, yielding a coherent theoretical picture in which memory decay, trajectory optimisation, and posterior concentration are manifestations of a single underlying generative mechanism.

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1 Introduction and Motivation

Modern large language models (LLMs) based on the transformer architecture generate text by iteratively sampling tokens from a conditional distribution. While the engineering details of these models are well understood, a unified *theoretical* framework that explains their behaviour through the lens of classical probability, control theory, and Bayesian statistics remains fragmented.

In this work, we propose that three classical formalisms—hidden Markov processes, stochastic optimal control, and Bayesian inference—each capture a complementary facet of transformer behaviour, and that together they form a coherent and mutually reinforcing picture.

Core thesis. Let $\boldsymbol{h}_t \in \mathbb{R}^d$ denote the hidden representation of a transformer at decoding step t, and let $x_t \in \mathcal{V}$ denote the emitted token. We argue:

(i) The stochastic process $\{(\boldsymbol{h}_t, x_t)\}_{t\geq 1}$ constitutes a *hidden Markov process* on a continuous state space with exponentially decaying memory.

(ii) Autoregressive generation solves a *stochastic optimal control problem* in which $\boldsymbol{h}_t$ is the state, x_t is the action, and the model has internalised an approximate value function.

(iii) The model parameters encode a *Bayesian prior* over sequences that is updated in-context via amortised approximate inference.

Notation. Throughout, we use $\mathcal{V}$ for the token vocabulary (finite, $|\mathcal{V}| = V$), d for the hidden dimension, T for the sequence length or horizon, $\boldsymbol{x}_{1:t} = (x_1, \ldots, x_t)$ for token prefixes, and $\boldsymbol{h}_t \in \mathbb{R}^d$ for the hidden state at step t. Probability measures are denoted $\mathbb{P}$, densities p, and expectations $\mathbb{E}$. Parameters of the model are collectively denoted $\theta \in \Theta$.

2 Preliminaries

2.1 Markov Chains on General State Spaces

Definition 2.1 (Markov kernel)**.** Let $(\mathcal{S}, \mathcal{B}(\mathcal{S}))$ be a measurable space. A *Markov kernel* (or transition kernel) is a map $\kappa : \mathcal{S} \times \mathcal{B}(\mathcal{S}) \to [0, 1]$ such that (i) for each $s \in \mathcal{S}$, $\kappa(s, \cdot)$ is a probability measure on $\mathcal{B}(\mathcal{S})$; and (ii) for each $B \in \mathcal{B}(\mathcal{S})$, $\kappa(\cdot, B)$ is measurable.

Definition 2.2 (Markov chain on a general state space)**.** A sequence of random variables $\{S_t\}_{t\geq 0}$ taking values in $\mathcal{S}$ is a *Markov chain* with kernel κ if for all t and all $B \in \mathcal{B}(\mathcal{S})$,

$$\mathbb{P}(S_{t+1} \in B \mid S_0, \ldots, S_t) = \kappa(S_t, B) \quad \text{a.s.}$$

Definition 2.3 (Ergodicity and mixing)**.** A Markov chain with kernel κ and stationary distribution π is *geometrically ergodic* if there exists $\rho \in (0, 1)$ and a measurable function $M : \mathcal{S} \to [0, \infty)$ such that for all $s \in \mathcal{S}$ and $t \geq 0$,

$$\|\kappa^t(s, \cdot) - \pi(\cdot)\|_{\mathrm{TV}} \leq M(s)\, \rho^t,$$

where $\|\cdot\|_{\mathrm{TV}}$ denotes the total variation norm and κ^t is the t-fold composition of κ.

The quantity $\tau_{\mathrm{mix}} = \lceil -1/\log \rho \rceil$ is the *mixing time*: beyond this horizon, the chain's distribution is ε-close to π regardless of the initial state.

2.2 Hidden Markov Models

Definition 2.4 (Hidden Markov Model)**.** A *Hidden Markov Model* (HMM) is a tuple $(\mathcal{S}, \mathcal{X}, \kappa, g, \nu)$ where:

- $\mathcal{S}$ is the (possibly continuous) hidden state space,
- $\mathcal{X}$ is the observation space,
- κ is a Markov transition kernel on $\mathcal{S}$,
- $g : \mathcal{S} \to \mathcal{P}(\mathcal{X})$ is the emission distribution,
- $\nu \in \mathcal{P}(\mathcal{S})$ is the initial distribution.

The generative model factorises as

$$s_0 \sim \nu, \qquad s_{t+1} \sim \kappa(s_t, \cdot), \qquad x_t \sim g(s_t).$$

2.3 Stochastic Optimal Control

Definition 2.5 (Discrete-time stochastic control problem)**.** A *stochastic optimal control problem* is specified by:

- A state space $\mathcal{S}$, action space $\mathcal{A}$,
- State dynamics $s_{t+1} \sim f(s_t, a_t, \xi_t)$ where ξ_t is i.i.d. noise,
- A per-step reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$,
- A terminal reward $R_T : \mathcal{S} \to \mathbb{R}$,
- A discount factor $\gamma \in (0, 1]$.

The objective is to find a policy $\pi : \mathcal{S} \to \mathcal{P}(\mathcal{A})$ maximising

$$J(\pi) = \mathbb{E}^{\pi}\left[\sum_{t=0}^{T} \gamma^t \, r(s_t, a_t) + \gamma^T R_T(s_T)\right].$$

Theorem 2.6 (Bellman optimality, discrete-time)**.** *The optimal value function $V^* : \mathcal{S} \to \mathbb{R}$ satisfies*

$$V^*(s) = \max_{a \in \mathcal{A}} \{ r(s, a) + \gamma \, \mathbb{E}_{s' \sim f(s,a,\cdot)}[V^*(s')] \}$$

and the optimal policy is $\pi^*(s) = \arg\max_a \{ r(s, a) + \gamma \, \mathbb{E}[V^*(f(s, a, \xi))] \}$.

2.4 Bayesian Inference

Definition 2.7 (Bayesian framework)**.** Given a prior $p(\theta)$ over parameters and a likelihood $p(\mathcal{D}|\theta)$ for data $\mathcal{D}$, the posterior is

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)\, p(\theta)}{p(\mathcal{D})}, \qquad p(\mathcal{D}) = \int p(\mathcal{D}|\theta)\, p(\theta)\, d\theta.$$

In the sequel, we will use the Bayesian framework not over model parameters θ (which are fixed at inference time) but over *sequence continuations*: the model parameters define a prior over token sequences, and the observed context serves as evidence.

3 The Transformer as a Hidden Markov Process

3.1 Defining the Hidden State

Consider a transformer with parameters θ operating autoregressively. At decoding step t, the model has consumed tokens $\boldsymbol{x}_{1:t}$ and produced an internal representation that we denote $\boldsymbol{h}_t \in \mathbb{R}^d$. Formally, $\boldsymbol{h}_t$ comprises the residual stream activations and, in cached inference, the key-value (KV) cache:

$$\boldsymbol{h}_t = \Phi_\theta(\boldsymbol{x}_{1:t}),$$

where $\Phi_\theta : \mathcal{V}^{\leq T} \to \mathbb{R}^d$ is the deterministic encoding map defined by the transformer's forward pass.

Proposition 3.1 (Markov property of the hidden state)**.** *The next-token distribution depends on* $\boldsymbol{x}_{1:t}$ only through $\boldsymbol{h}_t$*:*

$$p_\theta(x_{t+1} \mid \boldsymbol{x}_{1:t}) = p_\theta(x_{t+1} \mid \boldsymbol{h}_t) = \mathrm{softmax}\big(W_{\mathrm{out}}\, \boldsymbol{h}_t^{(L)}\big)_{x_{t+1}},$$

where $\boldsymbol{h}_t^{(L)}$ *is the final-layer representation and* $W_{\mathrm{out}} \in \mathbb{R}^{V \times d}$ *is the unembedding matrix. Hence the pair process* $\{(\boldsymbol{h}_t, x_t)\}_{t \geq 1}$ *is Markovian with transition kernel*

$$\kappa_\theta\big((\boldsymbol{h}_t, x_t),\, B \times \{v\}\big) = p_\theta(v \mid \boldsymbol{h}_t) \cdot \mathbf{1}\big[\Phi_\theta(\boldsymbol{x}_{1:t}, v) \in B\big] \tag{1}$$

for $B \in \mathcal{B}(\mathbb{R}^d)$ *and* $v \in \mathcal{V}$*.*

Proof. Since $\boldsymbol{h}_{t+1} = \Phi_\theta(\boldsymbol{x}_{1:t}, x_{t+1})$ is a deterministic function of $(\boldsymbol{h}_t, x_{t+1})$ (via the incremental KV-cache update), and x_{t+1} is sampled from $p_\theta(\cdot \mid \boldsymbol{h}_t)$, the state at $t+1$ depends on the past only through $(\boldsymbol{h}_t, x_t)$. The emission x_{t+1} depends only on $\boldsymbol{h}_t$. This establishes the Markov property. □

Remark 3.2 (Non-trivial Markov structure)*.* One might object that this is "trivially" Markovian since $\boldsymbol{h}_t$ deterministically encodes the full history $\boldsymbol{x}_{1:t}$. However, the claim becomes substantive when we consider the *effective* memory of the process—i.e., the degree to which distant tokens actually influence future predictions through the hidden state.

3.2 The HMM Interpretation

Definition 3.3 (Transformer HMM)**.** The *Transformer Hidden Markov Model* (T-HMM) is the tuple $(\mathcal{S}, \mathcal{V}, \kappa_\theta, g_\theta, \nu_\theta)$ where:

- $\mathcal{S} = \mathbb{R}^d$ is the continuous hidden state space,
- $\mathcal{V}$ is the finite token vocabulary (observation space),
- κ_θ is the kernel from (1),
- $g_\theta(\boldsymbol{h}) = \mathrm{softmax}(W_{\mathrm{out}}\, \boldsymbol{h}^{(L)})$ is the emission distribution,
- ν_θ is the distribution over initial hidden states induced by the start-of-sequence token.

Unlike classical HMMs with finite discrete hidden states, the T-HMM has a continuous, high-dimensional hidden space with *deterministic* state transitions conditioned on the emitted token. This places it in the family of *deterministic-transition HMMs*, closely related to predictive state representations and observable operator models.

3.3 Exponential Decay of Influence

The central empirical and theoretical observation is that the influence of token x_s on the prediction at step $t \gg s$ decays with the lag $t - s$.

Definition 3.4 (Influence function)**.** The *influence* of token position s on the prediction at position t is

$$I(s,t) \triangleq \mathbb{E}\Big[\mathrm{KL}\Big(p_\theta(\cdot \mid \boldsymbol{x}_{1:t}) \;\Big\|\; p_\theta(\cdot \mid \boldsymbol{x}_{1:t}^{\setminus s})\Big)\Big],$$

where $\boldsymbol{x}_{1:t}^{\setminus s}$ denotes the sequence with token x_s replaced by a draw from the marginal $p_\theta(x_s)$.

Theorem 3.5 (Exponential influence decay)**.** *Suppose the hidden-state transition map* $F_\theta : \mathbb{R}^d \times \mathcal{V} \to \mathbb{R}^d$ *defined by* $\boldsymbol{h}_{t+1} = F_\theta(\boldsymbol{h}_t, x_{t+1})$ *is Lipschitz in its first argument with constant* $\lambda < 1$ *uniformly over* $\mathcal{V}$*:*

$$\|F_\theta(\boldsymbol{h}, v) - F_\theta(\boldsymbol{h}', v)\| \leq \lambda \, \|\boldsymbol{h} - \boldsymbol{h}'\| \quad \forall\, v \in \mathcal{V},\ \boldsymbol{h}, \boldsymbol{h}' \in \mathbb{R}^d.$$

Then the influence decays geometrically:

$$I(s,t) \leq C\,\lambda^{2(t-s)}$$

for a constant C depending on the Lipschitz constant of g_θ and the diameter of the hidden state space.

Proof sketch. A perturbation $\delta \boldsymbol{h}_s = \boldsymbol{h}_s - \boldsymbol{h}'_s$ at step s propagates forward as

$$\|\delta \boldsymbol{h}_t\| \leq \lambda^{t-s}\,\|\delta \boldsymbol{h}_s\|$$

by iterated application of the contraction. Since g_θ is Lipschitz (the softmax composed with a linear map), the KL divergence between the resulting emission distributions is bounded by $O(\|\delta \boldsymbol{h}_t\|^2)$ via Pinsker's inequality and a local quadratic bound on KL, giving the $\lambda^{2(t-s)}$ rate. □ □

Remark 3.6 (Connection to mixing). In the language of Definition 2.3, the T-HMM is geometrically ergodic with rate $\rho = \lambda^2$ whenever the contraction condition holds. The mixing time $\tau_{\text{mix}} = O(1/|\log \lambda|)$ determines the effective memory horizon of the process.

Remark 3.7 (Attention as non-uniform contraction). In practice, multi-head attention can selectively break the contraction for specific positions (e.g., syntactic anchors, proper nouns), creating *non-uniform* decay. The process is better modelled as a *time-inhomogeneous* Markov chain with position-dependent contraction rates λ_t. The cumulative decay over a window $[s,t]$ is then $\prod_{\tau=s}^{t-1} \lambda_\tau$, which is still exponentially small when $\bar{\lambda} = (\prod \lambda_\tau)^{1/(t-s)} < 1$ on average.

3.4 Long-Run Memorylessness

Corollary 3.8 (Asymptotic memorylessness)**.** *Under the contraction condition of Theorem 3.5, for any $\varepsilon > 0$ there exists $\tau(\varepsilon)$ such that for all $t - s > \tau(\varepsilon)$,*

$$I(s,t) < \varepsilon.$$

In the limit $t \to \infty$ (with stationarity), the process is memoryless in the long run*: the predictive distribution converges to a fixed point independent of arbitrarily distant history.*

This formalises the intuition: at every step, the process has *local memory*—the current state $\boldsymbol{h}_t$ fully determines the next transition. But the influence of any particular historical token on that state is *vanishingly small* beyond the mixing horizon. The process is neither memoryless in the white-noise sense (where the present carries no information) nor long-memory in the fractional Brownian motion sense. It occupies the intermediate regime of *exponentially decaying, finite-horizon memory.*

4 Autoregressive Generation as Optimal Control

We now reinterpret the generative process through the lens of stochastic optimal control.

4.1 The Control Problem Formulation

Definition 4.1 (Autoregressive control problem)**.** Define the *autoregressive control problem* (ACP) as follows:

- **State space:** $\mathcal{S} = \mathbb{R}^d$ (hidden representations).
- **Action space:** $\mathcal{A} = \mathcal{V}$ (token vocabulary).
- **Dynamics:** $\boldsymbol{h}_{t+1} = F_\theta(\boldsymbol{h}_t, x_{t+1})$ (deterministic given action).
- **Policy:** $\pi_\theta(\cdot \mid \boldsymbol{h}_t) = g_\theta(\boldsymbol{h}_t) \in \mathcal{P}(\mathcal{V})$.
- **Reward:** To be specified by the training objective.
- **Horizon:** T (sequence length or end-of-sequence).

4.2 Pretraining as Maximum Likelihood Control

Under maximum likelihood training, the implicit reward is the log-probability of the observed next token:

$$r_{\mathrm{MLE}}(\boldsymbol{h}_t, x_{t+1}) = \log p_\theta(x_{t+1} \mid \boldsymbol{h}_t).$$

The pretraining objective maximises the cumulative reward over the data distribution:

$$J_{\mathrm{MLE}}(\theta) = \mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}}\left[\sum_{t=0}^{T-1} \log p_\theta(x_{t+1} \mid \boldsymbol{h}_t)\right] = \mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}}[\log p_\theta(\boldsymbol{x})].$$

Proposition 4.2 (MLE as undiscounted control)**.** *Maximum likelihood pretraining is equivalent to solving the ACP with $\gamma = 1$, $r(\boldsymbol{h}_t, a) = \log p_\theta(a \mid \boldsymbol{h}_t)$, and $R_T = 0$, under the constraint that the policy matches the model's own conditional distribution.*

This is a *self-consistent* control problem: the reward function *is* the policy's own log-probability, and training adjusts θ until the policy is optimal under its own induced reward.

4.3 RLHF as Reward Shaping

Reinforcement Learning from Human Feedback introduces an external reward signal. Let $R_\phi : \mathcal{V}^T \to \mathbb{R}$ denote a learned reward model with parameters ϕ, trained on human preference data.

Definition 4.3 (RLHF objective)**.** The RLHF objective is

$$J_{\mathrm{RLHF}}(\theta) = \mathbb{E}_{\boldsymbol{x}\sim\pi_\theta}\big[R_\phi(\boldsymbol{x})\big] - \beta\,\mathrm{KL}\big(\pi_\theta \| \pi_{\mathrm{ref}}\big), \tag{2}$$

where π_{ref} is the pretrained (reference) policy and $\beta > 0$ controls the strength of the KL penalty.

Theorem 4.4 (Optimal RLHF policy)**.** *The policy maximising* (2) *has the Gibbs form*

$$\pi^*_\theta(\boldsymbol{x}) \propto \pi_{\mathrm{ref}}(\boldsymbol{x})\,\exp\big(\beta^{-1} R_\phi(\boldsymbol{x})\big). \tag{3}$$

Proof. This is the standard result from the variational characterisation of the KL-regularised optimisation. The functional $\mathcal{L}[\pi] = \mathbb{E}_\pi[R_\phi] - \beta\,\mathrm{KL}(\pi\|\pi_{\mathrm{ref}})$ is strictly concave in π and its unique maximiser is the tilted distribution (3). □ □

Remark 4.5 (Control interpretation)*.* In the control framework, RLHF replaces the self-referential MLE reward with a *trajectory-level* reward $R_\phi(\boldsymbol{x}_{1:T})$, and the KL penalty acts as a *control cost* penalising deviation from the reference dynamics. The optimal policy (3) is precisely the solution to the KL-regularised stochastic control problem, a well-known result in linearly-solvable MDPs and path-integral control.

4.4 The Partition Function and the Free-Boundary Bellman Equation

The key insight is that RLHF rewards *entire sequences*, not individual tokens. A model that must emit tokens sequentially but is evaluated on trajectory-level reward is *forced* to decompose that trajectory reward into per-step decisions. Moreover, the horizon T is not fixed: the model decides *when to stop* by emitting a distinguished end-of-sequence token $\langle\mathrm{eos}\rangle \in \mathcal{V}$. This makes T endogenous—the stopping time is part of the solution, not a given.

Definition 4.6 (Stopping time)**.** The *generation stopping time* is the random variable

$$\tau = \inf\{t \geq 1 : x_t = \langle\mathrm{eos}\rangle\},$$

i.e., the first time the model emits the end-of-sequence token. The completed sequence is $\boldsymbol{x}_{1:\tau}$, and the trajectory reward is $R_\phi(\boldsymbol{x}_{1:\tau})$.

Definition 4.7 (Trajectory partition function with endogenous stopping)**.** For a hidden state $\boldsymbol{h}_t$ at step t, define the *trajectory partition function*

$$Z(\boldsymbol{h}_t) \triangleq \mathbb{E}_{\pi_{\text{ref}}}\left[\exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:\tau})\big) \;\middle|\; \boldsymbol{h}_t\right], \tag{4}$$

where the expectation is over all continuations under π_{ref}, including the stochastic stopping time $\tau \geq t$.

Theorem 4.8 (Free-boundary backward recursion)**.** *The partition function satisfies the backward recursion*

$$Z(\boldsymbol{h}_t) = \underbrace{\pi_{\text{ref}}(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) \cdot \exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t})\big)}_{S(\boldsymbol{h}_t)\ :\ \text{stopping value}} + \underbrace{\sum_{v \neq \langle\text{eos}\rangle} \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot Z\big(F_\theta(\boldsymbol{h}_t, v)\big)}_{C(\boldsymbol{h}_t)\ :\ \text{continuation value}}. \tag{5}$$

The stopping value $S(\boldsymbol{h}_t)$ is the expected exponentiated reward from terminating now; the continuation value $C(\boldsymbol{h}_t)$ is the expected exponentiated reward from emitting one more non-terminal token and proceeding optimally thereafter.

Proof. By the law of total expectation, conditioning on whether the next token is $\langle\text{eos}\rangle$ or not:

$$\begin{aligned}
Z(\boldsymbol{h}_t) &= \mathbb{E}_{\pi_{\text{ref}}}\left[\exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:\tau})\big) \;\middle|\; \boldsymbol{h}_t\right] \\
&= \pi_{\text{ref}}(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) \cdot \exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t})\big) \\
&\quad + \sum_{v \neq \langle\text{eos}\rangle} \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot \mathbb{E}_{\pi_{\text{ref}}}\left[\exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:\tau})\big) \;\middle|\; \boldsymbol{h}_{t+1} = F_\theta(\boldsymbol{h}_t, v)\right] \\
&= S(\boldsymbol{h}_t) + C(\boldsymbol{h}_t).
\end{aligned}$$

□

□

Remark 4.9 (The free-boundary structure). Equation (5) is structurally different from the fixed-horizon Bellman equation. In the fixed-horizon case, the terminal condition is given at a known T. Here, the "terminal condition" $\exp(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t}))$ is available at *every* step—the model can always choose to stop. The recursion simultaneously determines:

(a) The value function $V^*(\boldsymbol{h}_t) = \beta \log Z(\boldsymbol{h}_t)$.

(b) The optimal stopping region $\mathcal{S}^* = \{\boldsymbol{h} : S(\boldsymbol{h}) \geq C(\boldsymbol{h})\}$—the set of hidden states where terminating is preferred to continuing.

The stopping boundary $\partial\mathcal{S}^*$ is not given a priori; it is a *free boundary* that emerges from the recursion.

Definition 4.10 (Soft value function)**.** Define the *soft value function* as

$$V^*(\boldsymbol{h}_t) \triangleq \beta \log Z(\boldsymbol{h}_t). \tag{6}$$

Theorem 4.11 (Free-boundary soft Bellman equation)**.** *The soft value function satisfies*

$$\exp\big(\beta^{-1} V^*(\boldsymbol{h}_t)\big) = \pi_{\text{ref}}(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) \cdot \exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t})\big) + \sum_{v \neq \langle\text{eos}\rangle} \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot \exp\big(\beta^{-1} V^*\big(F_\theta(\boldsymbol{h}_t, v)\big)\big). \tag{7}$$

This is the free-boundary *soft Bellman equation: it is the Bellman equation of a linearly-solvable MDP with an endogenous stopping time.*

Proof. Immediate from Theorem 4.8 and (6): apply $\beta \log$ to both sides of (5). □ □

Remark 4.12 (Connection to optimal stopping theory). The recursion (7) is the *Snell envelope* of the process $\{R_\phi(\boldsymbol{x}_{1:t})\}_{t\geq 1}$ under the soft-max aggregation induced by β. In the limit $\beta \to 0$ (deterministic policy), it reduces to the classical optimal stopping rule:

$$V^*(\boldsymbol{h}_t) = \max\Big\{ \underbrace{R_\phi(\boldsymbol{x}_{1:t})}_{\text{stop}},\ \underbrace{\max_{v\neq\langle\text{eos}\rangle} V^*\big(F_\theta(\boldsymbol{h}_t, v)\big)}_{\text{continue}} \Big\},$$

which is the Bellman equation of an American option: exercise (emit EOS, collect R_ϕ) or continue (emit another token, reach a potentially higher-value state). The optimal stopping boundary $\partial\mathcal{S}^*$ is the *exercise boundary* of this option.

Remark 4.13 (Backward induction and lookahead). The endogenous stopping time forces the model to perform implicit backward induction. At each step, the model does not ask "what is the best complete response to the user's query?"—a planning problem requiring knowledge of the full trajectory. Instead it asks "what is the next token that brings the hidden state closer to the optimal stopping region while accumulating high value-to-go?" This is a *local* question answered by the gradient $\nabla_{\boldsymbol{h}} V^*(\boldsymbol{h}_t)$: the direction in hidden-state space along which the value function increases most steeply toward $\mathcal{S}^*$. The model surfs the value landscape toward the stopping boundary, one token at a time.

When this gradient is well-defined and smooth, local hill-climbing in advantage space produces globally coherent responses. When it is flat or noisy—the model has poor value estimates in that region of hidden space—the result is rambling, premature stopping, or inability to conclude. The model has lost the gradient toward $\partial\mathcal{S}^*$.

Theorem 4.14 (Optimal autoregressive policy via Z)**.** *The optimal policy for the RLHF objective* (2)*, with endogenous stopping, factorises autoregressively at each step as*

$$\pi^*(v \mid \boldsymbol{h}_t) = \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot \frac{Z\big(F_\theta(\boldsymbol{h}_t, v)\big)}{Z(\boldsymbol{h}_t)}, \qquad v \neq \langle\text{eos}\rangle, \tag{8}$$

and for the stopping action:

$$\pi^*(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) = \pi_{\text{ref}}(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) \cdot \frac{\exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t})\big)}{Z(\boldsymbol{h}_t)}. \tag{9}$$

The model emits EOS when $R_\phi(\boldsymbol{x}_{1:t})$ (the reward from stopping) is large relative to $V^*(F_\theta(\boldsymbol{h}_t, v))$ (the value from continuing).

Proof. The optimal joint distribution is $\pi^*(\boldsymbol{x}_{1:\tau}) \propto \pi_{\text{ref}}(\boldsymbol{x}_{1:\tau}) \exp(\beta^{-1} R_\phi(\boldsymbol{x}_{1:\tau}))$ by Theorem 4.4, where τ is the stopping time. Factoring autoregressively and using the definition of Z:

For $v \neq \langle\text{eos}\rangle$:

$$\begin{aligned}\pi^*(v \mid \boldsymbol{h}_t) &= \frac{\sum_{\tau>t} \sum_{\boldsymbol{x}_{t+2:\tau}} \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \pi_{\text{ref}}(\boldsymbol{x}_{t+2:\tau} \mid F_\theta(\boldsymbol{h}_t, v)) \exp(\beta^{-1} R_\phi(\boldsymbol{x}_{1:\tau}))}{Z(\boldsymbol{h}_t)} \\ &= \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot \frac{Z(F_\theta(\boldsymbol{h}_t, v))}{Z(\boldsymbol{h}_t)}. \end{aligned} \tag{10}$$

For $v = \langle\text{eos}\rangle$ (stopping at step $t+1$, so the sequence terminates as $\boldsymbol{x}_{1:t+1}$ with $x_{t+1} = \langle\text{eos}\rangle$):

$$\pi^*(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) = \frac{\pi_{\text{ref}}(\langle\text{eos}\rangle \mid \boldsymbol{h}_t) \cdot \exp(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t}))}{Z(\boldsymbol{h}_t)}. \tag{11}$$

One can verify that these sum to 1 by (5). □ □

Remark 4.15 (The Z-ratio as the central object). The ratio

$$\Gamma(\boldsymbol{h}_t, v) \triangleq \begin{cases} \dfrac{Z\big(F_\theta(\boldsymbol{h}_t, v)\big)}{Z(\boldsymbol{h}_t)} & v \neq \langle \text{eos} \rangle, \\ \dfrac{\exp\big(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t})\big)}{Z(\boldsymbol{h}_t)} & v = \langle \text{eos} \rangle, \end{cases} \tag{12}$$

is the single mathematical object that unifies all three frameworks, as we shall prove in Section 6. For continuation tokens, Γ measures the relative value of continuing via v. For EOS, Γ measures the relative value of stopping now versus continuing. The model emits EOS when $\Gamma(\boldsymbol{h}_t, \langle \text{eos} \rangle)$ dominates all $\Gamma(\boldsymbol{h}_t, v)$ for $v \neq \langle \text{eos} \rangle$.

Corollary 4.16 (The model must encode the value function)**.** *Since the RLHF-trained model approximates π^*, and π^* factorises through Z via (8)–(9), the model's hidden state $\boldsymbol{h}_t$ must encode sufficient information to approximate both $V^*(\boldsymbol{h}_t) = \beta \log Z(\boldsymbol{h}_t)$ (the continuation value) and $R_\phi(\boldsymbol{x}_{1:t})$ (the stopping value). The residual stream carries the log-partition function of the value-weighted future* and *the current accumulated reward.*

Remark 4.17 (Chain-of-thought as stopping-time extension). Chain-of-thought (CoT) prompting delays the stopping decision by emitting intermediate reasoning tokens. In the free-boundary framework, CoT shifts the optimal stopping region: the reasoning tokens navigate the hidden state to a region where $C(\boldsymbol{h}_t) \gg S(\boldsymbol{h}_t)$ and then eventually to a region where $S(\boldsymbol{h}_t)$ is large and $R_\phi(\boldsymbol{x}_{1:t})$ has been increased by the intervening reasoning. The model is given more time to climb the value landscape before being forced to exercise the option.

4.5 Pontryagin vs. Dynamic Programming

The dual formulations of optimal control illuminate different aspects of transformer behaviour.

Proposition 4.18 (Gradient as costate)**.** *During backpropagation through time (BPTT), the gradient of the loss with respect to the hidden state at time t,*

$$\boldsymbol{\lambda}_t = \frac{\partial \mathcal{L}}{\partial \boldsymbol{h}_t},$$

satisfies the adjoint equation

$$\boldsymbol{\lambda}_t = \frac{\partial r_t}{\partial \boldsymbol{h}_t} + \left(\frac{\partial F_\theta}{\partial \boldsymbol{h}_t}\right)^\top \boldsymbol{\lambda}_{t+1}, \tag{13}$$

which is the discrete-time Pontryagin costate equation. The gradient $\boldsymbol{\lambda}_t$ decays backward with rate governed by the spectral radius of $\partial F_\theta / \partial \boldsymbol{h}_t$—the same quantity controlling forward influence decay (Section 3.3).

Remark 4.19 (Symmetry of forward and backward decay). The contraction rate λ in Theorem 3.5 governs both:

(a) *Forward*: decay of influence of past tokens on future predictions.
(b) *Backward*: decay of gradient signal (vanishing gradients in BPTT).

This is not a coincidence: it is the primal-dual symmetry of optimal control, where the forward dynamics and backward adjoint share the same Jacobian structure.

5 The Bayesian Interpretation

5.1 Pretrained Weights as a Prior Measure

Definition 5.1 (Sequence prior)**.** The pretrained model p_θ defines a probability measure μ_θ over $\mathcal{V}^* = \bigcup_{t=1}^{\infty} \mathcal{V}^t$ (the space of all finite token sequences, where termination occurs at the first

$\langle$eos$\rangle$ emission) via the autoregressive factorisation:

$$\mu_\theta(\boldsymbol{x}_{1:\tau}) = \prod_{t=1}^{\tau} p_\theta(x_t \mid \boldsymbol{x}_{1:t-1}), \tag{14}$$

where $\tau = \inf\{t : x_t = \langle\text{eos}\rangle\}$ is the stopping time. We call μ_θ the *pretrained prior*. It is a measure over variable-length sequences with endogenous termination.

Proposition 5.2 (Embedding space as prior geometry)**.** *The token embedding matrix $W_{\text{emb}} \in \mathbb{R}^{d\times V}$ induces a geometry on $\mathcal{V}$ via the inner product*

$$\langle v, w\rangle_{\text{emb}} = \boldsymbol{e}_v^\top \boldsymbol{e}_w,$$

where $\boldsymbol{e}_v = W_{\text{emb}}[\cdot, v]$. This geometry encodes the prior's belief about semantic relatedness: tokens that are substitutable in many contexts (and hence have similar conditional distributions under μ_θ) are embedded nearby. Formally, if $\mathrm{KL}(p_\theta(\cdot|x_t = v)\|p_\theta(\cdot|x_t = w))$ is small for most contexts, then $\|\boldsymbol{e}_v - \boldsymbol{e}_w\|$ tends to be small.

This establishes that the embedding space is not merely a lookup table but a *learned prior geometry* encoding distributional similarity.

5.2 RLHF as Prior Reweighting

Theorem 5.3 (RLHF posterior)**.** *The optimal RLHF policy (3) has a Bayesian interpretation:*

$$\underbrace{\pi^*(\boldsymbol{x})}_{\textit{posterior}} \propto \underbrace{\mu_\theta(\boldsymbol{x})}_{\textit{prior}} \cdot \underbrace{\exp\!\big(\beta^{-1} R_\phi(\boldsymbol{x})\big)}_{\textit{likelihood (value)}}. \tag{15}$$

The reward model R_ϕ plays the role of a log-likelihood, and β controls the strength of the evidence relative to the prior. As $\beta \to \infty$, $\pi^* \to \mu_\theta$ (prior dominates); as $\beta \to 0$, π^* concentrates on $\arg\max R_\phi$ (likelihood dominates).

Remark 5.4 (Three layers of prior structure)*.* The full Bayesian hierarchy in a deployed conversational model comprises:

1. **Pretrained prior** μ_θ: broad beliefs about linguistic plausibility, learned from the training corpus. Encodes *what can follow what.*
2. **RLHF-shaped prior** π^*_θ: the prior reweighted by human values. Encodes *what should follow what* in general.
3. **In-context posterior** $\pi^*_\theta(\cdot \mid \boldsymbol{c})$: the RLHF prior further updated by the conversation context $\boldsymbol{c}$. Encodes *what should follow what for this user, now.*

5.3 In-Context Learning as Amortised Bayesian Inference

Definition 5.5 (In-context posterior)**.** Given a conversation context $\boldsymbol{c} = (c_1, \ldots, c_m)$ comprising the user's messages and the model's prior responses, the *in-context posterior* over continuations $\boldsymbol{x}_{m+1:T}$ is

$$p_\theta(\boldsymbol{x}_{m+1:T} \mid \boldsymbol{c}) = \frac{\mu_\theta(\boldsymbol{c},\, \boldsymbol{x}_{m+1:T})}{\mu_\theta(\boldsymbol{c})}. \tag{16}$$

Theorem 5.6 (ICL as amortised variational inference)**.** *The transformer's forward pass implements an* amortised *approximation to Bayesian updating. Specifically, the hidden state $\boldsymbol{h}_m = \Phi_\theta(\boldsymbol{c})$ serves as a* sufficient statistic *for the posterior over continuations:*

$$p_\theta(\boldsymbol{x}_{m+1:T} \mid \boldsymbol{c}) = p_\theta(\boldsymbol{x}_{m+1:T} \mid \boldsymbol{h}_m).$$

The attention mechanism computes the posterior by (approximately) evaluating the evidence integral

$$p_\theta(\boldsymbol{c}) = \sum_{\boldsymbol{z}} p_\theta(\boldsymbol{c} \mid \boldsymbol{z})\, p_\theta(\boldsymbol{z}),$$

where $\boldsymbol{z}$ ranges over latent "task" or "user" representations, and the summation is approximated by the feed-forward network's learned amortisation.

Remark 5.7 (Conversational Bayesian updating). In a multi-turn conversation, each user message c_t serves as new evidence that updates the posterior:

$$p_\theta(\boldsymbol{z} \mid c_1, \ldots, c_t) \propto p_\theta(c_t \mid \boldsymbol{z})\, p_\theta(\boldsymbol{z} \mid c_1, \ldots, c_{t-1}). \tag{17}$$

The model is inferring a latent representation $\boldsymbol{z}$ of the user's intent, expertise level, communication style, and desired outcomes. This posterior concentrates over the course of the conversation, explaining why model responses become more tailored as the dialogue progresses.

5.4 Posterior Concentration and Prompt Engineering

Proposition 5.8 (Prompt specificity controls posterior width)**.** *Let $\boldsymbol{c}$ be a prompt and let $\mathcal{H}_\theta(\boldsymbol{c})$ denote the entropy of the predictive distribution:*

$$\mathcal{H}_\theta(\boldsymbol{c}) = -\sum_{v \in \mathcal{V}} p_\theta(v \mid \boldsymbol{c}) \log p_\theta(v \mid \boldsymbol{c}).$$

More specific prompts yield lower entropy (tighter posteriors):

$$\boldsymbol{c} \sqsubseteq \boldsymbol{c}' \quad \Longrightarrow \quad \mathbb{E}[\mathcal{H}_\theta(\boldsymbol{c}')] \leq \mathbb{E}[\mathcal{H}_\theta(\boldsymbol{c})],$$

where $\sqsubseteq$ denotes that $\boldsymbol{c}'$ is an extension (refinement) of $\boldsymbol{c}$, and the expectation is over the randomness of the extension. This follows directly from the information-theoretic data processing inequality.

Corollary 5.9 (Prompt engineering as likelihood design)**.** *Prompt engineering is the craft of constructing a context $\boldsymbol{c}$ that serves as an effective likelihood function: one that concentrates the posterior $p_\theta(\cdot \mid \boldsymbol{c})$ onto the desired region of sequence space. A vague prompt is a weak likelihood (flat, uninformative); a precise prompt is a strong likelihood (peaked, informative).*

6 Unification: The Trinity

We now prove that the three perspectives developed in Sections 3–5 are not merely analogies but are *mathematically identical* descriptions of the same object: the partition-function ratio $\Gamma(\boldsymbol{h}_t, v)$ from (12).

6.1 The Pivotal Object

Recall from Theorem 4.14 that the optimal RLHF policy at each step takes the form $\pi^*(v \mid \boldsymbol{h}_t) = \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot \Gamma(\boldsymbol{h}_t, v)$ for all $v \in \mathcal{V}$ (including $\langle\text{eos}\rangle$), where Γ is defined by (12). The following theorem establishes that Γ is simultaneously the operative object in all three frameworks—and that the endogenous stopping decision is naturally embedded in each.

6.2 The Central Equivalence

Theorem 6.1 (The Trinity)**.** *Let π_{ref} be a pretrained autoregressive policy with hidden states $\boldsymbol{h}_t = \Phi_\theta(\boldsymbol{x}_{1:t})$, let $R_\phi : \mathcal{V}^* \to \mathbb{R}$ be a trajectory-level reward defined on variable-length sequences, and let π^* be the optimal policy of the KL-regularised objective (2) with endogenous stopping. Then the ratio $\Gamma(\boldsymbol{h}_t, v)$ from (12) admits three equivalent characterisations:*

(I) Optimal Control: Exponentiated Advantage with Stopping.
For continuation tokens $v \neq \langle\text{eos}\rangle$:

$$\Gamma(\boldsymbol{h}_t, v) = \exp\big(\beta^{-1} A^*(\boldsymbol{h}_t, v)\big), \tag{18}$$

where the soft advantage function *is $A^*(\boldsymbol{h}_t, v) = V^*(F_\theta(\boldsymbol{h}_t, v)) - V^*(\boldsymbol{h}_t)$ and $V^* = \beta \log Z$ satisfies the free-boundary soft Bellman equation (7).*

For the stopping action:

$$\Gamma(\boldsymbol{h}_t, \langle\text{eos}\rangle) = \exp\big(\beta^{-1}[R_\phi(\boldsymbol{x}_{1:t}) - V^*(\boldsymbol{h}_t)]\big). \tag{19}$$

The model stops when the immediate reward $R_\phi(\boldsymbol{x}_{1:t})$ exceeds the value-to-go $V^*(\boldsymbol{h}_t)$: the option is exercised when the current payoff dominates the expected future payoff.

(II) Bayesian Inference: Sequential Likelihood Ratio with Stopping as Commitment.

$$\Gamma(\boldsymbol{h}_t, v) = \frac{\pi^*(v \mid \boldsymbol{h}_t)}{\pi_{\text{ref}}(v \mid \boldsymbol{h}_t)} = \frac{\textit{posterior predictive}}{\textit{prior predictive}}, \tag{20}$$

for all $v \in \mathcal{V}$ (including $\langle\text{eos}\rangle$). For continuation tokens, Γ measures how much choosing v shifts the marginal likelihood of producing a high-value trajectory. For $v = \langle\text{eos}\rangle$, Γ measures the posterior probability that the evidence is sufficient*: stopping corresponds to the Bayesian decision that no further tokens would improve the expected reward. This is the* sequential probability ratio test *(SPRT) of Wald: the model continues gathering "evidence" (emitting tokens) until the posterior concentrates enough to commit.*

(III) Hidden Markov Model: Emission-Reweighted Transition with Absorbing State.

$$\Gamma(\boldsymbol{h}_t, v) = \frac{\kappa^*\big((\boldsymbol{h}_t, x_t),\, F_\theta(\boldsymbol{h}_t, v) \times \{v\}\big)}{\kappa_{\text{ref}}\big((\boldsymbol{h}_t, x_t),\, F_\theta(\boldsymbol{h}_t, v) \times \{v\}\big)}, \tag{21}$$

where κ^* is the transition kernel of the hidden Markov process under π^* and κ_{ref} is the kernel under π_{ref}. For $v = \langle\text{eos}\rangle$, the transition leads to the absorbing state*: the HMM is killed. The optimal HMM reweights the absorption probability relative to the reference HMM by $\Gamma(\boldsymbol{h}_t, \langle\text{eos}\rangle)$, increasing it when stopping is advantageous and decreasing it when continuation is preferred.*

Proof. We prove each characterisation and their mutual equivalence.

Proof of (I). For continuation tokens $v \neq \langle\text{eos}\rangle$, by definition (6):

$$\Gamma(\boldsymbol{h}_t, v) = \frac{Z(F_\theta(\boldsymbol{h}_t, v))}{Z(\boldsymbol{h}_t)} = \frac{\exp(\beta^{-1} V^*(F_\theta(\boldsymbol{h}_t, v)))}{\exp(\beta^{-1} V^*(\boldsymbol{h}_t))} = \exp\big(\beta^{-1} A^*(\boldsymbol{h}_t, v)\big).$$

For $v = \langle\text{eos}\rangle$:

$$\Gamma(\boldsymbol{h}_t, \langle\text{eos}\rangle) = \frac{\exp(\beta^{-1} R_\phi(\boldsymbol{x}_{1:t}))}{\exp(\beta^{-1} V^*(\boldsymbol{h}_t))} = \exp\big(\beta^{-1}[R_\phi(\boldsymbol{x}_{1:t}) - V^*(\boldsymbol{h}_t)]\big).$$

The stopping advantage $R_\phi(\boldsymbol{x}_{1:t}) - V^*(\boldsymbol{h}_t)$ is positive precisely on the optimal stopping region $\mathcal{S}^*$: the model stops when the immediate reward exceeds the continuation value. The Bellman

decomposition—including the stopping boundary—is forced by the structure of the problem: trajectory-level reward with sequential token emission and endogenous termination. □

Proof of (II). From Theorem 4.14, $\pi^*(v \mid \boldsymbol{h}_t) = \pi_{\text{ref}}(v \mid \boldsymbol{h}_t) \cdot \Gamma(\boldsymbol{h}_t, v)$ for all $v \in \mathcal{V}$, so

$$\Gamma(\boldsymbol{h}_t, v) = \frac{\pi^*(v \mid \boldsymbol{h}_t)}{\pi_{\text{ref}}(v \mid \boldsymbol{h}_t)}.$$

Now, $\pi^*(\boldsymbol{x}_{1:\tau}) \propto \pi_{\text{ref}}(\boldsymbol{x}_{1:\tau}) \exp(\beta^{-1} R_\phi(\boldsymbol{x}_{1:\tau}))$ is a Bayesian posterior with prior π_{ref} and log-likelihood $\beta^{-1} R_\phi$ (Theorem 5.3).

For continuation tokens, Γ is the per-step likelihood ratio: how much choosing v updates the belief that the trajectory will be high-value. For $v = \langle\text{eos}\rangle$, Γ is the *commitment ratio*: it measures the posterior belief that the accumulated evidence (tokens emitted so far) is sufficient to achieve high reward without further generation. This is precisely the stopping rule of the *sequential probability ratio test* (Wald, 1945): continue sampling (emitting tokens) until the accumulated evidence crosses a threshold, then commit (emit EOS).

The sequential factorisation extends to variable-length sequences:

$$\frac{\pi^*(\boldsymbol{x}_{1:\tau})}{\pi_{\text{ref}}(\boldsymbol{x}_{1:\tau})} = \prod_{t=1}^{\tau} \Gamma(\boldsymbol{h}_{t-1}, x_t), \tag{22}$$

where the final factor $\Gamma(\boldsymbol{h}_{\tau-1}, \langle\text{eos}\rangle)$ encodes the stopping decision. The trajectory-level Bayes factor decomposes as a product of per-step likelihood ratios, with the number of factors itself being a random variable determined by the stopping rule. □

Proof of (III). The transition kernel of the T-HMM (Definition 3.3) under policy π is (cf. (1)):

$$\kappa_\pi\big((\boldsymbol{h}_t, x_t),\, B \times \{v\}\big) = \pi(v \mid \boldsymbol{h}_t) \cdot \mathbf{1}\big[F_\theta(\boldsymbol{h}_t, v) \in B\big].$$

For $v \neq \langle\text{eos}\rangle$, the state transition F_θ is deterministic given $(\boldsymbol{h}_t, v)$, and the Radon-Nikodým derivative reduces to the emission ratio:

$$\left.\frac{d\kappa^*}{d\kappa_{\text{ref}}}\right|_{(\boldsymbol{h}_t, x_t),\, v} = \frac{\pi^*(v \mid \boldsymbol{h}_t)}{\pi_{\text{ref}}(v \mid \boldsymbol{h}_t)} = \Gamma(\boldsymbol{h}_t, v).$$

For $v = \langle\text{eos}\rangle$, the transition leads to an *absorbing state* $\boldsymbol{h}_\dagger$. The Radon-Nikodým derivative of the absorption probability is

$$\frac{\pi^*(\langle\text{eos}\rangle \mid \boldsymbol{h}_t)}{\pi_{\text{ref}}(\langle\text{eos}\rangle \mid \boldsymbol{h}_t)} = \Gamma(\boldsymbol{h}_t, \langle\text{eos}\rangle).$$

The optimal HMM is obtained from the reference HMM by tilting *both* the continuation emissions and the absorption probability by Γ. □

Mutual consistency. The three characterisations are three readings of the single identity, valid for all $v \in \mathcal{V}$ including $\langle\text{eos}\rangle$:

$$\boxed{\underbrace{\exp\big(\beta^{-1} A^*(\boldsymbol{h}_t, v)\big)}_{\text{(I) advantage / stopping value}} = \underbrace{\frac{\pi^*(v|\boldsymbol{h}_t)}{\pi_{\text{ref}}(v|\boldsymbol{h}_t)}}_{\text{(II) likelihood ratio / SPRT}} = \underbrace{\left.\frac{d\kappa^*}{d\kappa_{\text{ref}}}\right|_{(\boldsymbol{h}_t, x_t),\, v}}_{\text{(III) kernel tilt / absorption}} = \Gamma(\boldsymbol{h}_t, v).} \tag{23}$$

All three are *defined* as the same ratio from (12) and therefore agree identically, not approximately. When $v = \langle\text{eos}\rangle$, the identity connects the optimal stopping decision across all three frameworks: the exercise boundary (control), the commitment threshold (Bayesian), and the absorption rate (HMM) are the same surface in hidden-state space. □ □

6.3 The Necessity of Value Encoding

Theorem 6.2 (Structural necessity of Bellman decomposition)**.** *Let π_θ be any autoregressive policy that approximates the optimal RLHF policy π^* in the sense that $\mathrm{KL}(\pi^*\|\pi_\theta) \le \varepsilon$. Then there exists a function $\hat{V} : \mathbb{R}^d \to \mathbb{R}$ such that:*

(i) $\hat{V}$ is computable from the hidden state: $\hat{V}(\boldsymbol{h}_t) = \psi(\boldsymbol{h}_t)$ for some function ψ extractable from the model's parameters.

(ii) $\hat{V}$ satisfies the soft Bellman equation to within $O(\sqrt{\varepsilon})$:

$$\left|\hat{V}(\boldsymbol{h}_t) - \beta \log \mathbb{E}_{v\sim\pi_{\mathrm{ref}}}\left[\exp\!\big(\beta^{-1}\hat{V}(F_\theta(\boldsymbol{h}_t, v))\big)\right]\right| \le O(\sqrt{\varepsilon}).$$

Proof. Define $\hat{V}(\boldsymbol{h}_t) = \beta \log \sum_{v\in\mathcal{V}} \pi_\theta(v|\boldsymbol{h}_t)/\pi_{\mathrm{ref}}(v|\boldsymbol{h}_t) \cdot \pi_{\mathrm{ref}}(v|\boldsymbol{h}_t) \cdot \exp(\beta^{-1}\hat{V}(F_\theta(\boldsymbol{h}_t, v)))$—i.e., the soft value function *induced by* π_θ as if it were optimal. Since $\pi_\theta \approx \pi^*$ (in KL), we have $\pi_\theta(v|\boldsymbol{h}_t)/\pi_{\mathrm{ref}}(v|\boldsymbol{h}_t) \approx \Gamma(\boldsymbol{h}_t, v)$ pointwise in a total-variation sense (by Pinsker's inequality, $\|\pi_\theta(\cdot|\boldsymbol{h}_t) - \pi^*(\cdot|\boldsymbol{h}_t)\|_{\mathrm{TV}} \le \sqrt{\varepsilon/2}$). Substituting into the soft Bellman equation and bounding the residual via the Lipschitz continuity of log on $[p_{\min}, 1]$ yields the $O(\sqrt{\varepsilon})$ bound. □ □

Remark 6.3 (Interpretation)*.* Theorem 6.2 says: you cannot be a good RLHF policy without approximately solving the Bellman equation. The Bellman structure is not an optional interpretation—it is a *necessary consequence* of optimising a trajectory-level reward through sequential token emission. The model has no choice but to encode an approximate value function.

6.4 Decomposition of the Global Bayes Factor

Corollary 6.4 (Telescoping identity with endogenous stopping)**.** *The trajectory-level Bayes factor (posterior-to-prior ratio) decomposes as a telescoping product of per-step Γ-ratios, including the stopping decision:*

$$\frac{\pi^*(\boldsymbol{x}_{1:\tau})}{\pi_{\mathrm{ref}}(\boldsymbol{x}_{1:\tau})} = \frac{\exp(\beta^{-1}R_\phi(\boldsymbol{x}_{1:\tau}))}{Z_0} = \prod_{t=1}^{\tau}\Gamma(\boldsymbol{h}_{t-1}, x_t) = \exp\left(\beta^{-1}\sum_{t=1}^{\tau} A^*(\boldsymbol{h}_{t-1}, x_t)\right), \tag{24}$$

where $x_\tau = \langle\mathrm{eos}\rangle$ and the final factor $\Gamma(\boldsymbol{h}_{\tau-1}, \langle\mathrm{eos}\rangle)$ encodes the stopping decision. This identity says three things simultaneously:

(a) ***Control:*** *The trajectory reward decomposes into a sum of per-step advantages* including the stopping advantage*—this is the reward decomposition lemma of the free-boundary Bellman framework. The number of terms is itself optimised.*

(b) ***Bayesian:*** *The full Bayes factor decomposes into a product of per-step likelihood ratios, with the final factor being the* commitment decision*—the chain rule of Bayesian updating with an endogenous sample size (SPRT).*

(c) ***HMM:*** *The Radon-Nikodým derivative of the trajectory measure decomposes into per-step kernel ratios terminating at absorption—the Girsanov-type change of measure for Markov chains with killing.*

These three decompositions are the same identity read in three languages. The endogenous stopping time τ is the same in all three: the exercise boundary, the commitment threshold, and the absorption time are one object.

Proof. For the telescoping product, noting that for $t < \tau$, $x_t \ne \langle\mathrm{eos}\rangle$ so $\Gamma(\boldsymbol{h}_{t-1}, x_t) = Z(\boldsymbol{h}_t)/Z(\boldsymbol{h}_{t-1})$, and for $t = \tau$, $x_\tau = \langle\mathrm{eos}\rangle$ so $\Gamma(\boldsymbol{h}_{\tau-1}, \langle\mathrm{eos}\rangle) = \exp(\beta^{-1}R_\phi(\boldsymbol{x}_{1:\tau-1}))/Z(\boldsymbol{h}_{\tau-1})$:

$$\begin{aligned}\prod_{t=1}^{\tau}\Gamma(\boldsymbol{h}_{t-1}, x_t) &= \left(\prod_{t=1}^{\tau-1}\frac{Z(\boldsymbol{h}_t)}{Z(\boldsymbol{h}_{t-1})}\right)\cdot\frac{\exp(\beta^{-1}R_\phi(\boldsymbol{x}_{1:\tau-1}))}{Z(\boldsymbol{h}_{\tau-1})}\\ &= \frac{Z(\boldsymbol{h}_{\tau-1})}{Z(\boldsymbol{h}_0)}\cdot\frac{\exp(\beta^{-1}R_\phi(\boldsymbol{x}_{1:\tau-1}))}{Z(\boldsymbol{h}_{\tau-1})} = \frac{\exp(\beta^{-1}R_\phi(\boldsymbol{x}_{1:\tau}))}{Z_0}.\end{aligned} \tag{25}$$

The telescoping cancellation is exact, and the stopping factor merges seamlessly. □ □

6.5 Summary: Why the Trinity Holds

The Trinity is not a coincidence, a metaphor, or an approximation. It holds because:

1. **Sequential emission + trajectory reward + endogenous stopping ⇒ free-boundary Bellman.** Any agent that must act step-by-step, is evaluated on the cumulative outcome, and decides when to stop is solving a free-boundary dynamic programming problem. The soft Bellman equation with stopping (Theorem 4.11) is the *unique* decomposition of the trajectory-level RLHF objective into per-step decisions plus a stopping rule.
2. **Gibbs form + chain rule + endogenous sample size ⇒ sequential testing.** The optimal RLHF policy has the Gibbs form (Theorem 4.4), which IS Bayes' rule. Its autoregressive factorisation IS sequential Bayesian updating. The endogenous stopping time makes this a sequential probability ratio test: the model continues until the evidence is sufficient. The partition function $Z(\boldsymbol{h}_t)$ IS the marginal likelihood.
3. **Markov state + deterministic transitions + absorbing state ⇒ killed HMM.** The hidden state is Markov by construction (Proposition 3.1). The EOS token induces an absorbing state, creating a Markov chain with killing. The change from reference to optimal policy is a Radon-Nikodým tilt of both the emission kernel and the absorption rate.
4. **All three meet at Γ, including at the boundary.** The ratio Γ is the unique object that simultaneously encodes the per-step advantage, the per-step likelihood ratio, and the per-step kernel derivative. For $v \neq \langle\text{eos}\rangle$, it governs continuation. For $v = \langle\text{eos}\rangle$, it governs stopping. The optimal stopping region—the exercise boundary, the commitment threshold, the absorption surface—is the same set $\mathcal{S}^*$ in all three frameworks.

6.6 Convergence of the Three Decay Mechanisms

A further consistency check: all three frameworks predict the same memory decay, governed by the same spectral parameter.

Proposition 6.5 (Unified decay)**.** *Under the contraction condition of Theorem 3.5:*

(a) ***HMM****: Influence of token x_s on prediction at t decays as $O(\lambda^{2(t-s)})$ (forward mixing).*

(b) ***Control****: The advantage $A^*(\boldsymbol{h}_t, v)$'s sensitivity to a perturbation at step s decays as $O(\lambda^{t-s})$, since A^* depends on $V^*(\boldsymbol{h}_t)$ which inherits the contraction of $\boldsymbol{h}_t$ (cf. (13)).*

(c) ***Bayesian****: The per-step likelihood ratio $\Gamma(\boldsymbol{h}_t, v)$'s sensitivity to evidence at step s decays as $O(\lambda^{2(t-s)})$ (posterior decorrelation).*

All three rates are governed by the same quantity λ: the contraction rate of the hidden-state dynamics F_θ.

7 Implications and Interpretive Consequences

7.1 Coherence Degradation in Long Generations

Proposition 7.1 (Ergodic drift)**.** *For generation lengths $T \gg \tau_{\text{mix}}$, the autoregressive process converges toward the stationary distribution of the Markov chain, losing coherence with the initial prompt. This is the long-run memorylessness described in Corollary 3.8: the "plot" set up in the opening tokens is gradually forgotten as the process mixes.*

This explains the empirical observation that very long model generations tend to become generic, repetitive, or drift from the original topic—even when the context window has not been exhausted.

7.2 Chain-of-Thought as Control Horizon Extension

In the control framework, CoT prompting expands the effective planning horizon.

Proposition 7.2. *Let T_0 be the base horizon and $T_{\mathrm{CoT}} = T_0 + K$ the horizon with K additional reasoning tokens. The value achievable satisfies*

$$V^*_{T_{\mathrm{CoT}}}(\boldsymbol{h}_0) \geq V^*_{T_0}(\boldsymbol{h}_0)$$

with strict inequality whenever the optimal trajectory under T_0 is suboptimal—i.e., whenever intermediate reasoning steps allow the controller to reach higher-value terminal states.

7.3 Few-Shot Learning as Strong Likelihood

In the Bayesian framework, few-shot examples act as concentrated evidence.

Proposition 7.3. *Let $\boldsymbol{c}_k$ be a prompt containing k input-output demonstrations. The posterior entropy satisfies*

$$\mathcal{H}_\theta(\boldsymbol{c}_k) \leq \mathcal{H}_\theta(\boldsymbol{c}_{k-1}) \leq \cdots \leq \mathcal{H}_\theta(\boldsymbol{c}_0),$$

with equality iff the additional examples are redundant given the existing context. Each example narrows the posterior, concentrating the model's predictive distribution on the task-relevant region of sequence space.

7.4 Temperature as Posterior Sharpening

Proposition 7.4. *Sampling with temperature $\tau > 0$ modifies the emission distribution to*

$$p_\tau(v \mid \boldsymbol{h}_t) = \frac{p_\theta(v \mid \boldsymbol{h}_t)^{1/\tau}}{\sum_w p_\theta(w \mid \boldsymbol{h}_t)^{1/\tau}}.$$

In the Bayesian framework, $\tau < 1$ corresponds to tempering *the posterior (increasing confidence beyond what the evidence warrants), while $\tau > 1$ corresponds to* flattening *(increased epistemic humility). In the control framework, τ modulates the* exploration-exploitation trade-off *of the stochastic policy.*

8 Formal Connections to Classical Theory

8.1 Relationship to Predictive State Representations

The T-HMM of Definition 3.3 is closely related to Predictive State Representations (PSRs). In a PSR, the state is defined as a vector of predictions about future observables, and the system is "observable" in the sense that the state can be maintained from observations alone.

Proposition 8.1. *The transformer hidden state $\boldsymbol{h}_t$ constitutes an* approximate *predictive state: it encodes (approximately) all information about the future predictive distribution that is extractable from the past observations $\boldsymbol{x}_{1:t}$. It differs from a classical PSR in that:*

(a) The state is computed by a learned, parametric function Φ_θ rather than maintained via an update rule derived from the system's dynamics.

(b) The approximation quality depends on model capacity and training adequacy.

8.2 Relationship to Information Bottleneck

Proposition 8.2. *The hidden state $\boldsymbol{h}_t$ solves an approximate* information bottleneck *problem:*

$$\min_{\boldsymbol{h}_t} I(\boldsymbol{x}_{1:t};\, \boldsymbol{h}_t) - \beta_{\mathrm{IB}}\, I(\boldsymbol{h}_t;\, x_{t+1}),$$

where $I(\cdot;\cdot)$ denotes mutual information. The state compresses the past ($\boldsymbol{x}_{1:t}$) while retaining information predictive of the future (x_{t+1}). The contraction rate λ controls how aggressively the bottleneck compresses distant information.

8.3 Relationship to Free Energy Principle

The RLHF objective (2) has a direct connection to variational free energy minimisation:

$$\begin{aligned} -J_{\mathrm{RLHF}}(\theta) &= -\mathbb{E}_{\pi_\theta}[R_\phi] + \beta\,\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}}) && (26)\\ &= \beta\big(-\beta^{-1}\mathbb{E}_{\pi_\theta}[R_\phi] + \mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})\big) && (27)\\ &= \beta\, \underbrace{\big(\mathbb{E}_{\pi_\theta}[-\log \tilde{p}(\boldsymbol{x})] + \mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})\big)}_{\text{variational free energy}}, && (28) \end{aligned}$$

where $\tilde{p}(\boldsymbol{x}) \propto \exp(\beta^{-1} R_\phi(\boldsymbol{x}))$. Minimising $-J_{\mathrm{RLHF}}$ is thus equivalent to minimising the variational free energy, connecting transformer training to the free energy principle from theoretical neuroscience.

9 The Spectral Structure of Memory

To make the decay analysis of Section 3.3 more precise, we study the spectral properties of the linearised hidden-state dynamics.

9.1 Transfer Operator

Definition 9.1 (Linearised transfer operator)**.** At a reference trajectory $\bar{\boldsymbol{h}}_{1:T}$, define the *transfer operator* at step t as

$$\mathcal{T}_t = \frac{\partial F_\theta(\bar{\boldsymbol{h}}_t, \bar{x}_{t+1})}{\partial \boldsymbol{h}_t} \in \mathbb{R}^{d\times d}.$$

The cumulative transfer from step s to step t is

$$\mathcal{T}_{s:t} = \prod_{\tau=s}^{t-1} \mathcal{T}_\tau.$$

Theorem 9.2 (Spectral characterisation of memory)**.** *Let $\sigma_1(\mathcal{T}_{s:t}) \geq \cdots \geq \sigma_d(\mathcal{T}_{s:t})$ be the singular values of $\mathcal{T}_{s:t}$. Then:*

(i) The effective rank *of the influence propagation from s to t is*

$$r_{\mathrm{eff}}(s,t) = \frac{\left(\sum_i \sigma_i\right)^2}{\sum_i \sigma_i^2} \leq d,$$

and this decreases with $|t-s|$ as modes decay at different rates.

(ii) The memory spectrum $\{\lambda_k\}_{k=1}^d$*, defined as the geometric means $\lambda_k = (\prod_\tau \sigma_k(\mathcal{T}_\tau))^{1/(t-s)}$, determines which directions in hidden space carry long-range information.*

(iii) Directions with $\lambda_k \approx 1$ correspond to persistent memory modes *(e.g., topic, style, factual anchors), while directions with $\lambda_k \ll 1$ carry only local information (e.g., recent syntactic context).*

This provides a mechanistic explanation for why transformers can simultaneously exhibit short-range syntactic sensitivity and long-range semantic coherence: the memory is *spectrally structured*, with different components decaying at different rates.

10 Discussion and Open Questions

10.1 Time Inhomogeneity

The T-HMM is fundamentally *time-inhomogeneous*: the effective transition kernel κ_θ varies with position (due to positional encodings) and with context (due to attention). Classical ergodic theory for time-homogeneous chains must be extended to handle this case. The relevant framework is that of *non-stationary Markov chains* with *uniform ergodicity* conditions.

10.2 Non-Monotonic Attention

The contraction assumption in Theorem 3.5 is violated when attention heads "reach back" to specific positions with high weight. This creates *punctuated memory*: the influence function $I(s,t)$ is not monotonically decreasing but exhibits spikes at positions of high attention. A more refined model would replace the uniform contraction with a *random* contraction rate, leading to a multiplicative random walk analysis.

10.3 Scaling Laws and the Prior

As model scale increases, the pretrained prior μ_θ becomes a better approximation to the "true" distribution over natural language. In the Bayesian framework, this means that larger models require *less* evidence (shorter prompts) to concentrate on the correct posterior—which is consistent with the empirical finding that larger models exhibit stronger few-shot learning.

10.4 Open Problems

1. **Exact characterisation of the spectral gap.** Can the mixing time of the T-HMM be bounded in terms of architectural parameters (depth, width, number of heads)?
2. **Finite-sample Bayesian consistency.** Under what conditions does the in-context posterior converge to the true task posterior as the number of in-context examples grows?
3. **Control complexity.** What is the computational complexity of the optimal control problem solved (approximately) by the transformer, and how does it relate to the model's inference cost?
4. **Value function geometry.** Can the implicit value function $V_\theta(\boldsymbol{h}_t)$ be extracted and analysed? Does it exhibit the smoothness properties assumed in approximate dynamic programming?
5. **Multi-agent control.** In conversational settings, the user and the model are both acting as controllers with coupled dynamics. The appropriate framework may be *Stackelberg games* or *cooperative control* rather than single-agent optimisation.

11 Conclusion

We have developed a unified theoretical framework that characterises autoregressive transformer language models through three equivalent lenses:

1. As **hidden Markov processes** with continuous-state dynamics and exponentially decaying memory, formalising the notion that these models are "memoryless in the long run" while retaining local, vanishingly influential memory at each step.
2. As **stochastic optimal controllers** that solve a Bellman equation over the hidden-state manifold, selecting tokens as control actions that steer the sequence toward high-value terminal states.
3. As **Bayesian inference engines** whose pretrained weights define a prior over language, RLHF reshapes this prior with a value-based likelihood, and in-context conditioning performs amortised posterior updating.

The three frameworks are not merely analogies but are provably equivalent formalisations, with the hidden state $\boldsymbol{h}_t$ playing the unifying role of Markov state, control state, and sufficient statistic simultaneously. The spectral structure of the hidden-state dynamics governs the decay of influence in all three frameworks, providing a single quantitative characterisation of the model's effective memory.

This unified perspective yields interpretive power: chain-of-thought prompting is control horizon extension; prompt engineering is likelihood design; RLHF is prior reweighting; temperature is posterior tempering; coherence degradation in long generations is ergodic mixing. These are not metaphors—they are mathematical consequences of the framework.

A Detailed Proof of Theorem 3.5

Proof. Let $\boldsymbol{x}_{1:t}$ be a token sequence and let $\boldsymbol{x}_{1:t}^{\setminus s}$ denote the sequence with x_s replaced by an independent draw $\tilde{x}_s \sim p_\theta(x_s)$. Define

$$\boldsymbol{h}_\tau = \Phi_\theta(\boldsymbol{x}_{1:\tau}), \qquad \tilde{\boldsymbol{h}}_\tau = \Phi_\theta(\boldsymbol{x}_{1:\tau}^{\setminus s}).$$

For $\tau < s$, $\boldsymbol{h}_\tau = \tilde{\boldsymbol{h}}_\tau$. At $\tau = s$, the perturbation introduces a discrepancy $\|\boldsymbol{h}_s - \tilde{\boldsymbol{h}}_s\| \leq D$ where D is the diameter of the embedding space.

For $\tau > s$, since $\boldsymbol{h}_{\tau+1} = F_\theta(\boldsymbol{h}_\tau, x_{\tau+1})$ and F_θ is λ-Lipschitz in its first argument:

$$\|\boldsymbol{h}_\tau - \tilde{\boldsymbol{h}}_\tau\| \leq \lambda^{\tau-s}\, D.$$

The emission distributions at step t are

$$p_\theta(\cdot \mid \boldsymbol{h}_t), \qquad p_\theta(\cdot \mid \tilde{\boldsymbol{h}}_t).$$

Since $g_\theta = \mathrm{softmax} \circ (W_{\mathrm{out}} \cdot)$ is Lipschitz with constant L_g (the softmax Jacobian is bounded by $1/4$ per coordinate, and W_{out} has bounded operator norm),

$$\|g_\theta(\boldsymbol{h}_t) - g_\theta(\tilde{\boldsymbol{h}}_t)\|_1 \leq L_g\, \|\boldsymbol{h}_t - \tilde{\boldsymbol{h}}_t\| \leq L_g\, D\, \lambda^{t-s}.$$

By Pinsker's inequality and the local quadratic bound on KL divergence:

$$\mathrm{KL}\big(g_\theta(\boldsymbol{h}_t) \| g_\theta(\tilde{\boldsymbol{h}}_t)\big) \leq \frac{\|g_\theta(\boldsymbol{h}_t) - g_\theta(\tilde{\boldsymbol{h}}_t)\|_1^2}{2\, \min_v p_\theta(v|\tilde{\boldsymbol{h}}_t)} \leq C'\, \lambda^{2(t-s)},$$

where $C' = L_g^2 D^2/(2p_{\min})$ and $p_{\min}$ is a lower bound on the emission probabilities (which exists for softmax with bounded logits).

Taking expectations over the randomness of $\tilde{x}_s$:

$$I(s,t) = \mathbb{E}\Big[\mathrm{KL}\big(g_\theta(\boldsymbol{h}_t) \| g_\theta(\tilde{\boldsymbol{h}}_t)\big)\Big] \leq C'\, \lambda^{2(t-s)} =: C\, \lambda^{2(t-s)}. \qquad \square$$

B Proof of Theorem 4.4

Proof. The RLHF objective is

$$J(\pi) = \mathbb{E}_\pi[R_\phi(\boldsymbol{x})] - \beta\,\mathrm{KL}(\pi\|\pi_{\mathrm{ref}}) = \sum_{\boldsymbol{x}} \pi(\boldsymbol{x})\left[R_\phi(\boldsymbol{x}) - \beta \log \frac{\pi(\boldsymbol{x})}{\pi_{\mathrm{ref}}(\boldsymbol{x})}\right].$$

Setting the functional derivative to zero:

$$\frac{\delta J}{\delta \pi(\boldsymbol{x})} = R_\phi(\boldsymbol{x}) - \beta \log \frac{\pi(\boldsymbol{x})}{\pi_{\mathrm{ref}}(\boldsymbol{x})} - \beta = 0.$$

Solving:

$$\log \pi^*(\boldsymbol{x}) = \log \pi_{\mathrm{ref}}(\boldsymbol{x}) + \beta^{-1} R_\phi(\boldsymbol{x}) - 1.$$

Exponentiating:

$$\pi^*(\boldsymbol{x}) \propto \pi_{\mathrm{ref}}(\boldsymbol{x})\, \exp\!\big(\beta^{-1} R_\phi(\boldsymbol{x})\big).$$

The proportionality constant is the normalising partition function $Z = \sum_{\boldsymbol{x}} \pi_{\mathrm{ref}}(\boldsymbol{x})\, \exp(\beta^{-1} R_\phi(\boldsymbol{x}))$. Strict concavity of J in π (since $-\mathrm{KL}$ is strictly concave) guarantees uniqueness. □ □

C Notation Reference

Symbol	Meaning
$\mathcal{V}$	Token vocabulary, $\lvert\mathcal{V}\rvert = V$
d	Hidden state dimension
T	Sequence length / control horizon
$\boldsymbol{x}_{1:t}$	Token sequence $(x_1, \ldots, x_t)$
$\boldsymbol{h}_t$	Hidden state at step t, $\boldsymbol{h}_t \in \mathbb{R}^d$
Φ_θ	Transformer encoding map
F_θ	Incremental hidden-state update
g_θ	Emission distribution (softmax output)
κ_θ	Markov transition kernel
μ_θ	Pretrained prior over sequences
π_θ	Policy (conditional token distribution)
R_ϕ	Learned reward model
V_θ	Value function (value-to-go)
λ	Contraction rate of hidden dynamics
β	KL penalty / inverse temperature
τ	Sampling temperature
τ_{mix}	Mixing time of the hidden Markov process